LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	14
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Full Authority Digital Control System
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Full Authority Digital Control System.
5. Specific Objectives	20.4.2.1 Trainee will be able to explain principles of Aero engine control systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of FCS Block Diagram and FADEC Logic Flow; Safety: Do not touch live avionics; Instructor supervision required for diagram analysis.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an engine control system?
(b) How did analog control systems differ from digital systems?
(c) Why is digital control preferred for modern aero engines?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Full Authority Digital Control • Comparison with Analog Control Systems • Role of FADEC in Engine Management • Sensors and Actuators in FCS • Benefits of Digital Over Analog Control	L & Ds	PPT, IB	15 min
(b)	• FADEC Architecture and Components • Engine Parameters Monitored by FCS • Control Logic and Decision Making • Fail-Safe Mechanisms in FCS • Integration with Aircraft Flight Control	L & Ds	PPT, PDF	20 min
(c)	• Advantages of FCS in Performance • Maintenance Implications of Digital Systems • Future Trends in Engine Control • Troubleshooting Common FCS Issues • Summary of FCS Principles	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between full authority and limited authority digital control.
(b) Describe how FADEC ensures engine safety during a sensor failure.
(c) List three advantages of digital control systems over analog systems.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of Full Authority Digital Control apply to the next lesson on Engine Vibration Monitoring Systems.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Purpose of Engine Vibration monitoring systems and Operation of Engine Vibration monitoring systems.	Refer to the master syllabus reference list	LP 15

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min